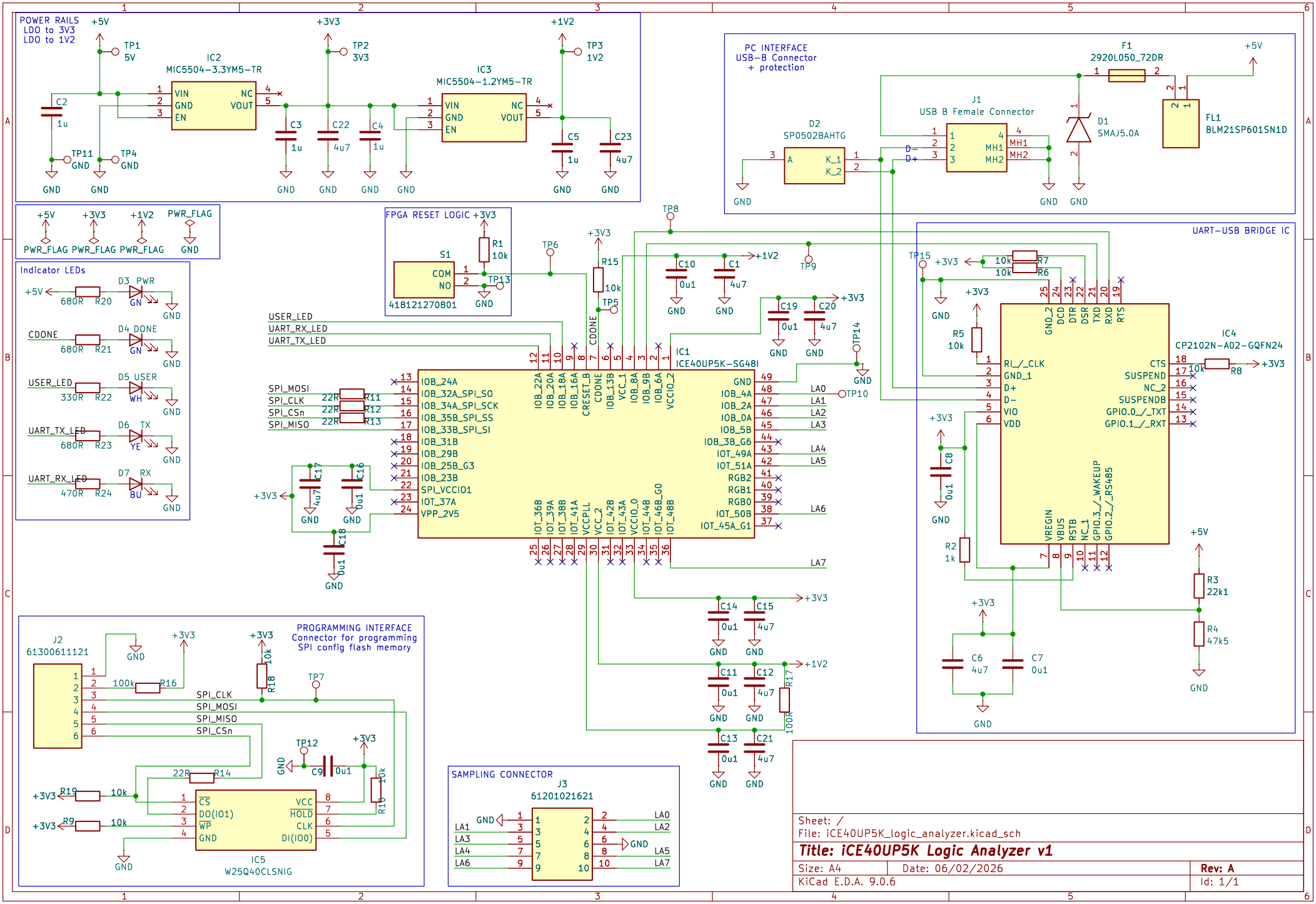
POWER RAILS
LDO to 3V3
LDO to 1V2
+5V
+3V3
+1V2
TP1
5V
TP2
3V3
TP3
1V2
IC2
MIC5504-3.3YM5-TR
IC3
MIC5504-1.2YM5-TR
VIN
GND
EN
NC
VOUT
C2
1u
C3
1u
C22
4u7
C4
1u
C5
1u
C23
4u7
TP11
GND
TP4
GND
PC INTERFACE
USB-B Connector
+ protection
F1
2920L050_72DR
J1
USB B Female Connector
D2
SP0502BAHTG
D1
SMAJ5.0A
FL1
BLM21SP601SN1D
D-
D+
MH1
MH2
PWR_FLAG
Indicator LEDs
D3_PWR
D4_DONE
D5_USER
D6_TX
D7_RX
680R
330R
470R
CDONE
USER_LED
UART_TX_LED
UART_RX_LED
FPGA RESET LOGIC
S1
418121270801
R1
10k
TP13
TP6
TP8
TP9
TP5
TP14
TP15
R15
10k
C10
0u1
C1
4u7
C19
C20
IC1
ICE40UP5K-SG48I
SPI_MOSI
SPI_CLK
SPI_CSn
SPI_MISO
22R
R11
R12
R13
C17
4u7
C16
0u1
C18
0u1
C14
C15
C11
C12
C13
C21
R17
100R
LA0
LA1
LA2
LA3
LA4
LA5
LA6
LA7
TP10
UART-USB BRIDGE IC
IC4
CP2102N-A02-GQFN24
R7
10k
R6
10k
R5
10k
R8
10k
R2
1k
C8
0u1
C6
4u7
C7
0u1
R3
22k1
R4
47k5
PROGRAMMING INTERFACE
Connector for programming
SPI config flash memory
J2
61300611121
R16
100k
R18
10k
TP7
TP12
C9
0u1
R14
22R
R19
10k
R9
10k
R10
10k
IC5
W25Q40CLSNIG
SAMPLING CONNECTOR
J3
61201021621
Sheet: /
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Title: iCE40UP5K Logic Analyzer v1
Size: A4
Date: 06/02/2026
Rev: A
KiCad E.D.A. 9.0.6
Id: 1/1